(a) fault and event library: 44 registered types (31 faults, 13 events)

Faults (31) — fault (abnormal condition) — event (real-world randomness)

Sensor (8)

- Bias — any sensor point
- Drift — any sensor point
- Gain error — any sensor point
- Excess noise — any sensor point
- Stuck value — any sensor point
- Spikes — any sensor point
- Dropout — any sensor point
- Override / spoofing — any sensor point

Actuator (7)

- VAV damper stuck — VAV
- Reheat stuck closed — VAV
- Reheat stuck open — VAV
- OA damper stuck — AHU
- CC valve stuck closed — AHU
- HC valve stuck closed — AHU
- Supply fan failure — AHU

Equipment (7)

- Chiller fouling — chiller
- Boiler fouling — boiler
- Tower fouling — cooling tower
- Filter loading — AHU
- Fan degradation — AHU
- Equipment trip — plant equipment
- Filter bypass — AHU

Control (8)

- Schedule always on — AHU
- Schedule never on — AHU
- PI mistuning — VAV, AHU
- Economizer disabled — AHU
- Economizer always on — AHU
- SAT reset disabled — AHU
- Controller frozen — VAV, AHU
- Set-point override — zone

Network (2)

- Communication loss — any equipment
- Historian outage — building

Events (13)

Occupancy (3)

- Occupancy surge — zone or building
- Meeting — zone
- Unscheduled holiday — building

Weather (3)

- Weather change — building
- Outdoor CO_2 change — building
- Wildfire smoke — building

Gains / envelope (3)

- Window open — zone
- Plug-load change — zone or building
- Lighting change — zone or building

Indoor air quality (1)

- Pollutant source — zone

Grid (2)

- Demand response — building
- Price / carbon spike — building

(b) severity profiles $s(t)$ of an injection; each type modulates one parameter, $\theta(t) = \theta_0 + s(t)\,\Delta\theta$

